

18. (3.5 points)

Given the following financial information for a P&C insurance company as of December 31, 2017:

- Firm's projected ROE = 12.0%.
- Projected dividend payout rate = 50.0%.
- Yield on long-term T-bonds = 3.0%.
- Liquidity premium for long-term T-bonds = 1.0%.
- Expected equity risk premium = 6.0%.
- Insurer's equity beta for the Company = 1.5.
- Insurer's book value as of December 31, 2017 = \$100M.

a. (1 point)

Calculate the company's Price to Book Value (P-BV) multiple, assuming projections continue into perpetuity.

b. (0.5 point)

Calculate the company's Price to Book Value (P-BV) multiple, assuming excess returns, if any, decline to the cost of capital after 3 years.

c. (0.5 point)

Identify and briefly discuss which Price to Book Value multiple would serve as a more reasonable indicator in the valuation of the company considering parts a. and b. above.

d. (0.5 point)

Describe one reason why transaction multiples based on initial public offerings or mergers and acquisitions of peer firms may be a better reflection in the valuation of the company than the approaches used in parts a. and b. above.

e. (1 point)

Describe two reasons why transaction multiples based on initial public offerings or mergers and acquisitions of peer firms may not be better reflections in the valuation of a company than in the approaches used in parts a. and b. above.

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SPRING 2018 EXAM 7, QUESTION 18	
TOTAL POINT VALUE: 3.5	LEARNING OBJECTIVE(S): B3
SAMPLE ANSWERS	
Part a: 1 point	
<u>Sample 1</u> Cost of Capital (k) = $(0.03 - 0.01) + 1.5 \times 0.06 = 0.11$ Growth Rate (g) = $(1 - \text{dividend payout rate}) \times \text{projected ROE} = (1 - 0.5) \times 0.12 = 0.06$ Price/BV = $1 + \frac{\text{ROE} - k}{k - g}$ = 1.2 <u>Sample 2</u> Risk-free rate r(f) = 3% – 2% = 1% Risk Premium = $E[r(m) - r(f)] = 6\%$ (given) Cost of Capital k = r(f) + beta × Risk Premium = 1% + 1.5 × 6% = 11% Growth rate g = Plowback ratio × ROE = $(1 - 50\%) \times 12\% = 6\%$ P-BV = $1 + (\text{ROE} - k) / (k - g) = 1 + (12\% - 11\%) / (11\% - 6\%) = 1.2$	
Part b: 0.5 point	
<u>Sample 1</u> Price/BV = $1 + [(\text{ROE} - k)/(k - g)] \times (1 - [(1 + g)/(1 + k)]^3)$ = $1 + [(0.12 - 0.11)/(0.11 - 0.06)] \times (1 - [(1 + 0.06)/(1 + 0.11)]^3)$ = 1.026 <u>Sample 2</u> Price/BV = $1 + \frac{(0.12 - 0.11)}{1.11} + \frac{(0.12 - 0.11) \times 0.66}{1.11^2} + \frac{(0.12 - 0.11) \times 0.33}{1.11^3}$ = 1.017	
Part c: 0.5 point	
<u>Sample 1</u> P-BV in part (b) is more reasonable since it assumes abnormal returns decline to zero as competitors enter the market to capture the excess returns. This is more reasonable. <u>Sample 2</u> The return horizon of 3 years of excess returns (in the part (b)) is more reasonable because over time competitors will enter the market and reduce profits.	
Part d: 0.5 point	
<u>Sample 1</u> Transaction multiples are based on complex negotiations between sophisticated parties. This is more meaningful than the market multiples in (a, b) that are more volatile.	

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Sample 2

Since the transaction multiples are obtained after complex negotiations between sophisticated parties, these multiples are more meaningful and less prone to random fluctuations.

Part e: 1 point

Sample responses:

- Control premiums – firms may overpay to acquire the target firm and control its operations.
- M&A overpricing – mergers and acquisitions lead to increase in the target firm's value, implying that the acquiring firm is overpaying.
- IPOs underpricing – IPOs tend to be underpriced, based on studies of historical transactions.

EXAMINER'S REPORT

Candidates were expected to calculate Price to Book Value (P-BV), understand the assumptions underlying the calculations, and know the advantages and disadvantages of market multiples vs. transaction multiples (IPOs and M&A).

Part a

Candidates were expected to calculate the Price to Book Value multiple given the set of financial information and assuming the abnormal earnings continued in perpetuity.

Common mistakes included:

- Miscalculating the cost of capital. Some candidates used the yield on long-term T-bonds (3.0%) as the risk-free rate in the CAPM equation, while others subtracted risk-free rate from the expected equity risk premium value that was given.
- Assuming that growth rate is 0%
- Using the wrong formula to calculate P-BV value

Part b

Candidates were expected to calculate the Price to Book Value assuming excess returns decline in 3 years.

Common mistakes included:

- Attempting to calculate a tail factor after 3 years. If the excess returns decline to zero, then there is no tail factor.
- Setting $k = \text{ROE}$ immediately in the first year, and calculating a resulting P-BV of 1.0.

Part c

Candidates were expected to identify that the P-BV in part b. would be a more reasonable indicator and explain that a company with abnormal earnings would see increased competition, making it unreasonable to earn abnormal earnings in perpetuity.

No credit was given to responses that identified the value in part a. as the most reasonable, because the company has a beta greater than 1.0 and, as a result, is taking risk and expecting to

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grow. A company with increased volatility cannot ensure positive abnormal earning in perpetuity.

Common mistakes included:

- Identifying that abnormal returns in perpetuity were unreasonable but failing to explain why.

Part d

Candidates were expected to explain that IPOs and M&A involve complex negotiations and sophisticated parties, making them more meaningful than market multiples.

Common mistakes included:

- Stating that transaction multiples are based on additional sources of data, as compared to the market multiples.

Part e

Candidates were expected to describe two reasons why multiples based on financial data may be better than multiples based on IPOs and M&A.

Common mistakes included:

- Reversing the fact that IPOs are usually underpriced and M&As are usually overpriced.
- Stating that IPOs are generally overpriced because investors are often overly optimistic of newly public companies.